

UCS503 - Software Engineering

PharmaFlowAI

An Intelligent Web-Based Management System for Retail
Pharmacy Operations

1024240056 **Takshit Mahajan**

1024240063 **Sahil Goyal**

1024240043 **Sidak Raj Viridi**

1024240059 **Ridhman Arora**

Group: 3X12 BE Third Year, CSE

Faculty Advisor

Dr. Anushka Chaudhary

Mid-Semester Evaluation

Computer Science and Engineering Department

Thapar Institute of Engineering and Technology, Patiala

September 8, 2026

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1 Introduction

1.1 Background and Context

Retail pharmacies manage several interconnected activities as part of their daily operations. These include maintaining medicine records, monitoring stock, managing batches, purchasing medicines from suppliers, maintaining customer records, processing sales, generating bills, and monitoring conditions such as low stock and approaching expiry.

These activities are closely connected. A medicine may have multiple batches with different quantities and expiry dates. A purchase increases available stock, while a sale decreases stock. Customer information can be associated with transactions, while historical sales information can be used to generate analytics.

PharmaFlowAI is developed as a centralized web-based retail pharmacy management system that brings these activities together in one platform. The application provides separate but connected modules for medicine management, inventory, suppliers, purchases, customers, billing, sales, notifications, and analytics.

The system is implemented using Python and the Django web framework. Django Templates together with HTML and CSS are used for the user interface, while Django models and the Django ORM provide the data-management foundation. Django Admin is also used for administrative data management during development.

The project also explores intelligent functionality. Machine Learning is currently being researched for possible applications such as demand forecasting, stock-out prediction, expiry-risk assessment, and intelligent reorder recommendations. These features are considered future intelligent extensions and are not presented as fully deployed ML models.

An AI chatbot has also been developed as a prototype. The chatbot is intended to provide a conversational method of accessing pharmacy-related information. Complete integration of the chatbot with operational application data is still pending because suitable project data is currently being generated and prepared.

1.2 Motivation

The primary motivation for PharmaFlowAI is to provide a centralized workspace for routine pharmacy operations. When medicine, inventory, purchasing, customer, and sales information is maintained through connected modules, the user can obtain a more consistent view of pharmacy activities.

Inventory management is particularly important in pharmacy operations. Medicines have different quantities, batches, reorder levels, and expiry dates. Therefore, simply maintaining a product catalogue is not sufficient. The system must also provide information that helps the user monitor stock and identify items requiring attention.

Another motivation is the use of data for analytics. Sales and inventory information can be summarized to provide useful operational indicators. In the future, historical information may also be used for predictive models.

The project therefore combines conventional pharmacy management functionality with a foundation for future AI and ML-based features.

1.3 Problem Statement

Retail pharmacies handle a large amount of interconnected operational information. Managing medicine records, inventory, suppliers, purchases, customers, billing, and sales separately can make it difficult to obtain a unified view of the pharmacy.

The problem addressed by PharmaFlowAI is the development of a centralized web-based platform capable of managing these major activities in an organized manner.

The system should allow the user to manage medicines and batches, monitor inventory, maintain supplier and purchase information, manage customers, record sales, generate billing information, view notifications, and analyze operational data.

A further objective is to investigate how intelligent features can be introduced into the system. The project therefore includes an AI chatbot prototype and research into suitable Machine Learning applications.

1.4 Scope of the Project

The scope of the current project includes the following functionality:

- User authentication and workspace access.
- Medicine catalogue management.
- Medicine search and filtering.

- Batch-level inventory management.
- Stock and reorder-level monitoring.
- Supplier management.
- Purchase management.
- Customer management.
- Billing and sales.
- Notifications.
- Descriptive analytics.
- AI chatbot prototype.
- Research for future ML-based analytics.

The current project does not claim production-ready ML prediction models. Similarly, the chatbot is currently a prototype and complete operational database integration is planned as future work.

1.5 Project Objectives

The major objectives of PharmaFlowAI are:

1. Develop a centralized Django-based pharmacy management system.
2. Maintain medicine information and batch-level inventory.
3. Connect suppliers and purchase information with pharmacy inventory.
4. Maintain customer records and sales transactions.
5. Provide billing and sales management.
6. Provide notifications for important inventory conditions.
7. Provide descriptive analytics based on operational information.
8. Develop an AI chatbot prototype.
9. Research suitable Machine Learning algorithms for future intelligent analytics.

1.6 Expected Outcomes

The expected outcome is a functional pharmacy management platform that provides a centralized workspace for major retail pharmacy operations.

The system should make operational information easier to manage and provide connections between medicines, inventory, purchasing, customers, and sales.

Another expected outcome is an operational data foundation for future analytics and Machine Learning. Historical records can potentially be used to identify demand patterns and inventory-related trends.

The chatbot prototype provides an initial step toward conversational interaction with the system.

1.7 Project Development Approach

The project follows a modular development approach.

The major stages are:

1. Identification of pharmacy requirements.
2. Definition of major system entities.
3. Design of UML and system architecture.
4. Development of core pharmacy modules.
5. Integration of inventory, purchasing, billing, and customer records.
6. Addition of notifications and analytics.
7. Development of the AI chatbot prototype.
8. Research and preparation for future ML integration.

This approach allows the core management system to be developed first while providing a foundation for intelligent functionality.

1.8 Organization of the Report

This report is divided into six major chapters.

Chapter 1 introduces PharmaFlowAI, its motivation, problem statement, objectives, and scope.

Chapter 2 presents the analysis phase, including stakeholders, use cases, functional requirements, non-functional requirements, user stories, and data requirements.

Chapter 3 presents the design phase, including system architecture, technical stack, UML diagrams, activity flow, data flow, database design, and security considerations.

Chapter 4 describes the implementation of the major system modules.

Chapter 5 discusses testing, evaluation, observed results, limitations, and the current status of AI and ML functionality.

Chapter 6 concludes the report and discusses future work and recommendations.

Appendices provide requirements traceability, testing information, and a module summary.

2 Analysis Phase

2.1 Stakeholders and Users

The primary user of PharmaFlowAI is the pharmacy owner or authorized pharmacy staff member.

The user interacts with the system to manage medicines, inventory, suppliers, purchases, customers, sales, billing, notifications, and analytics.

The system also contains supporting components such as Django Admin, analytics functionality, and the future AI/ML layer.

2.2 Operational Challenges

A pharmacy manages different categories of information.

Medicine records describe the products available in the pharmacy. Inventory records describe the available quantities and batches. Supplier and purchase records describe procurement. Customer and sales records describe transactions.

If these records are maintained separately, it can become difficult to obtain a unified operational view.

PharmaFlowAI addresses this issue by organizing the major activities within a connected web application.

2.3 Use Cases

The major use cases of the system include:

- Login and authentication.
- Dashboard access.
- Medicine management.
- Inventory management.

- Batch management.
- Supplier management.
- Purchase management.
- Customer management.
- Billing and sales.
- Notifications.
- Analytics.
- AI chatbot interaction.

2.3.1 Use Case Diagram

The Use Case Diagram represents the main interaction between the pharmacy owner and the PharmaFlowAI system.

The diagram provides a high-level representation of the functional capabilities available to the primary user.

2.3.2 Use Case Descriptions

Authentication

The authentication use case provides controlled access to the pharmacy workspace.

Dashboard

The dashboard provides a summary of important pharmacy information and acts as the primary navigation point.

Medicine Management

Medicine management allows the user to maintain medicine records including medicine name, generic name, strength, category, manufacturer, unit, prescription requirement, and active status.

Inventory Management

Inventory management provides information about available quantities, reorder levels, stock status, batches, shelves, and expiry-related conditions.

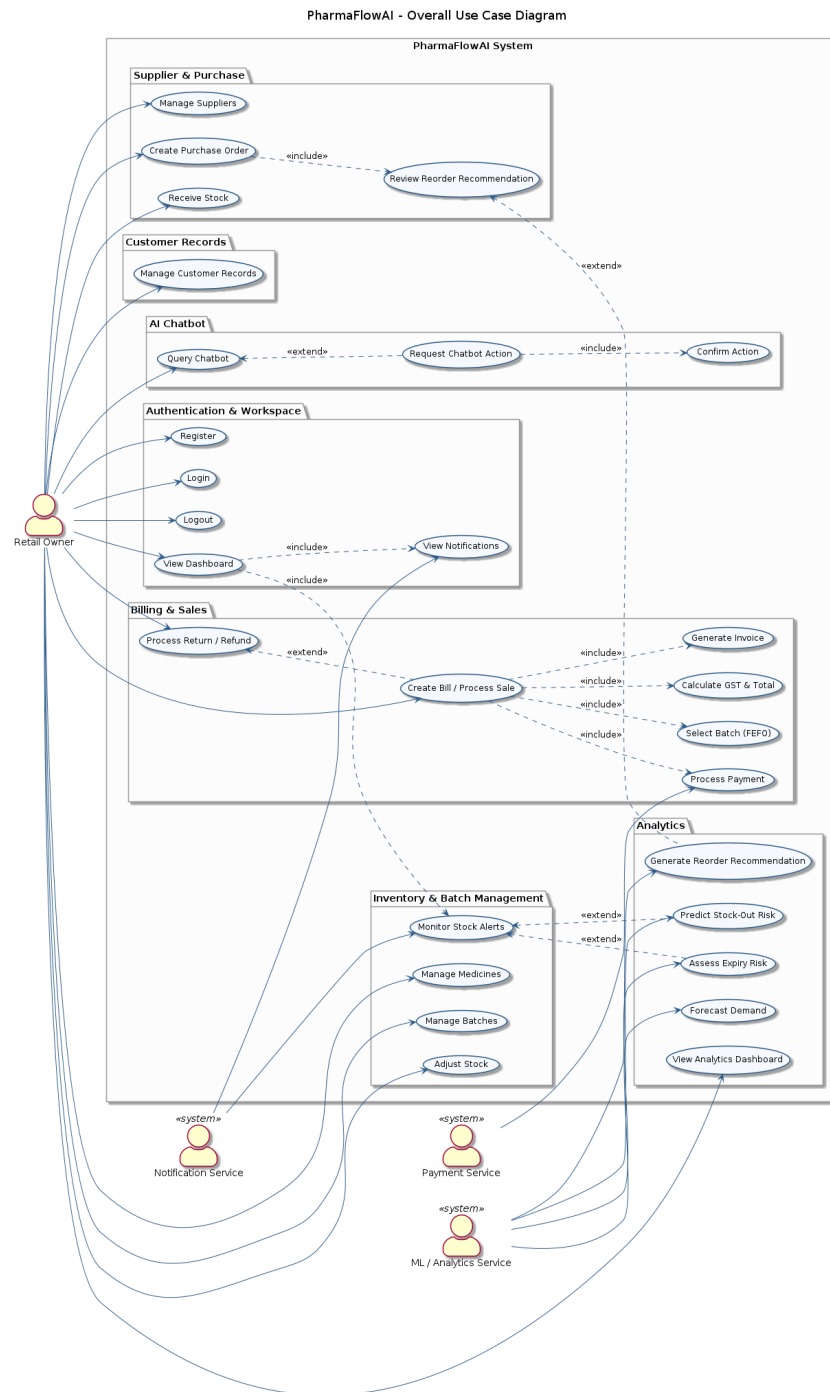


Figure 2.1: PharmaFlowAI Use Case Diagram

Supplier Management

Supplier management maintains supplier information required for procurement activities.

Purchase Management

Purchase management records procurement activity and represents incoming medicine stock.

Customer Management

Customer management maintains customer information associated with pharmacy transactions.

Billing and Sales

Billing and sales management records invoice information, customer details, purchased items, amount, and transaction status.

Notifications

Notifications highlight conditions that may require attention, such as low stock and approaching expiry.

Analytics

Analytics provides summarized information derived from operational records.

AI Chatbot

The chatbot provides a conversational interface for supported pharmacy-related questions.

2.4 Functional Requirements

2.4.1 Authentication Requirements

The system shall provide an authentication interface and restrict access to the main pharmacy workspace.

2.4.2 Medicine Requirements

The system shall allow users to:

- Add medicine records.
- Update medicine information.
- Search medicines.
- Filter medicines.
- Maintain categories.
- Maintain manufacturer information.
- Maintain medicine strength and unit.
- Specify prescription requirements.
- Maintain active or inactive status.

2.4.3 Inventory Requirements

The inventory module shall support:

- Batch-level tracking.
- Quantity monitoring.
- Reorder-level monitoring.
- Stock-status information.
- Shelf information.
- Expiry-related monitoring.

2.4.4 Supplier and Purchase Requirements

The system shall allow supplier records to be maintained and purchase information to be recorded.

2.4.5 Customer Requirements

The system shall maintain customer information that can be used during sales and billing.

2.4.6 Billing Requirements

The billing module shall support invoice number, customer, items, amount, and transaction status.

2.4.7 Notification Requirements

The system shall identify important inventory conditions such as low stock and approaching expiry.

2.4.8 Analytics Requirements

The analytics module shall provide summaries based on available operational information.

2.4.9 Chatbot Requirements

The chatbot shall provide conversational interaction for supported pharmacy questions and shall be extendable toward database-backed responses.

2.5 Non-Functional Requirements

2.5.1 Usability

The system should provide clear navigation and understandable interfaces for common pharmacy activities.

Search, filtering, status indicators, and dashboard summaries should help users locate information efficiently.

2.5.2 Maintainability

The Django-based structure should allow individual modules to be modified without redesigning the entire application.

2.5.3 Security

Authentication and controlled access are important because the application contains operational and customer-related information.

Detailed deployment-level security testing remains future work.

2.5.4 Scalability

The web-based design can be extended for larger datasets, additional users, or multiple pharmacy branches.

2.5.5 Extensibility

The architecture should allow analytics, ML, chatbot, and reporting functionality to be expanded later.

2.6 User Stories

User	User Story
Pharmacy Owner	As a pharmacy owner, I want to log into the system so that only authorized users can access the workspace.
Pharmacy Owner	As a pharmacy owner, I want to manage medicines so that the catalogue remains organized.
Pharmacy Owner	As a pharmacy owner, I want to search medicines so that records can be located efficiently.
Pharmacy Owner	As a pharmacy owner, I want to monitor stock so that low-stock medicines can be identified.
Pharmacy Owner	As a pharmacy owner, I want to monitor batches so that expiry information can be considered.
Pharmacy Owner	As a pharmacy owner, I want to manage suppliers and purchases so that procurement records remain organized.
Pharmacy Owner	As a pharmacy owner, I want to maintain customer information so that transactions can be associated with customers.
Pharmacy Owner	As a pharmacy owner, I want to create bills so that sales transactions are recorded.
Pharmacy Owner	As a pharmacy owner, I want notifications so that important stock conditions are visible.
Pharmacy Owner	As a pharmacy owner, I want analytics so that pharmacy performance can be summarized.
Pharmacy Owner	As a pharmacy owner, I want to use the chatbot so that supported questions can be asked conversationally.

2.7 Data Requirements

The major data areas are:

- Medicine information.
- Batch information.
- Inventory information.
- Supplier information.
- Purchase information.
- Customer information.
- Sales information.
- Payment information.
- Notification information.
- Analytics information.

Historical information is particularly important for future ML work.

Useful fields may include transaction dates, quantities, inventory levels, reorder levels, and expiry information.

2.8 Requirement Prioritization

The requirements can be grouped into three categories.

2.8.1 Core Requirements

Core requirements include authentication, medicine management, inventory, suppliers, purchases, customers, billing, and sales.

2.8.2 Supporting Requirements

Notifications and analytics provide additional monitoring and reporting capabilities.

2.8.3 Future Intelligent Requirements

The chatbot integration and ML-based predictions represent the intelligent extension of the system.

3 Design Phase

3.1 System Architecture

PharmaFlowAI follows a modular web architecture.

The user interacts with the web interface. Requests are handled by Django application logic, while application data is managed through Django models and the database.

The architecture separates the presentation layer, application layer, data layer, operational modules, analytics, and intelligent extensions.

3.1.1 High-Level Design

The high-level flow can be represented as:

User → Web Interface → Django Application → Django ORM / Database

Operational modules use this common application and data foundation.

3.1.2 Architecture Diagram

The architecture diagram shows the major layers and modules of PharmaFlowAI.

The architecture connects the user interface with Django application logic and the data layer. Operational modules form the main system, while analytics, the chatbot prototype, and ML research represent intelligent extensions.

3.2 Technical Stack

The technical stack of PharmaFlowAI consists of:

- **Programming Language:** Python.
- **Backend Framework:** Django.
- **Frontend:** Django Templates, HTML, and CSS.
- **Data Management:** Django Models and Django ORM.

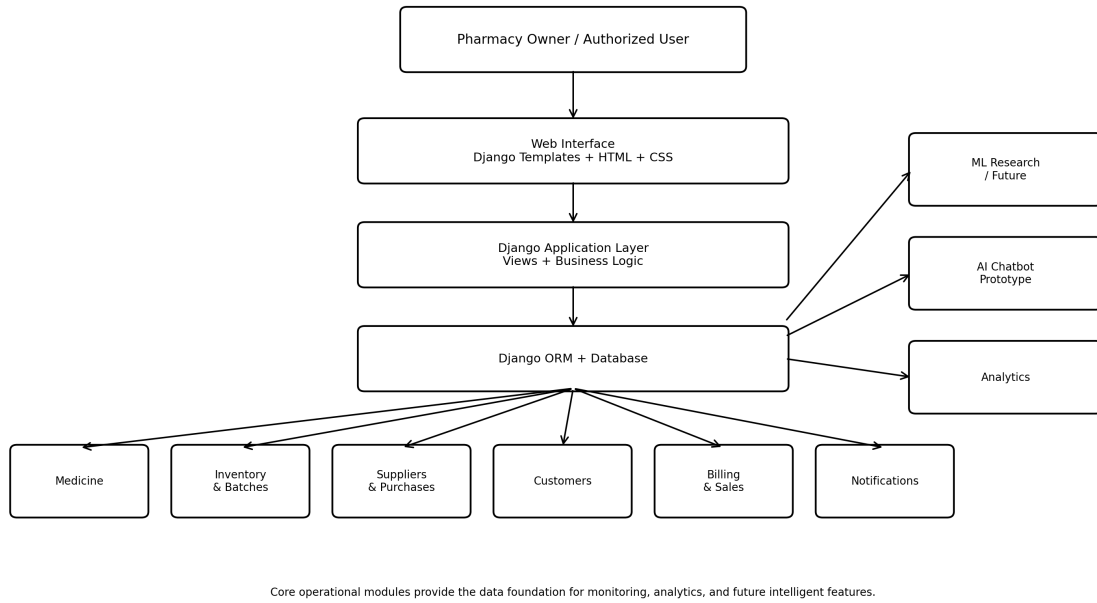


Figure 3.1: PharmaFlowAI High-Level System Architecture

- **Administrative Management:** Django Admin.
- **AI Component:** AI chatbot prototype.
- **ML Component:** Research and future integration.

3.3 System Modeling

System modeling represents the functional, structural, behavioral, and data-related aspects of PharmaFlowAI.

3.3.1 Class Diagram

The Class Diagram represents the structural organization of the application.

Major entities include Medicine, Batch, Supplier, PurchaseOrder, PurchaseItem, Sale, SaleItem, Customer, Payment, Notification, Recommendation, and Prediction.

The relationships between these entities support the connections between medicine management, inventory, procurement, billing, customers, and future intelligent functionality.

3.3.2 Activity Diagram

The Activity Diagram represents the flow of major pharmacy operations.

It begins with authentication and proceeds through medicine management, inventory updates, purchase decisions, customer purchases, billing, analytics, notifications, and chatbot interaction.

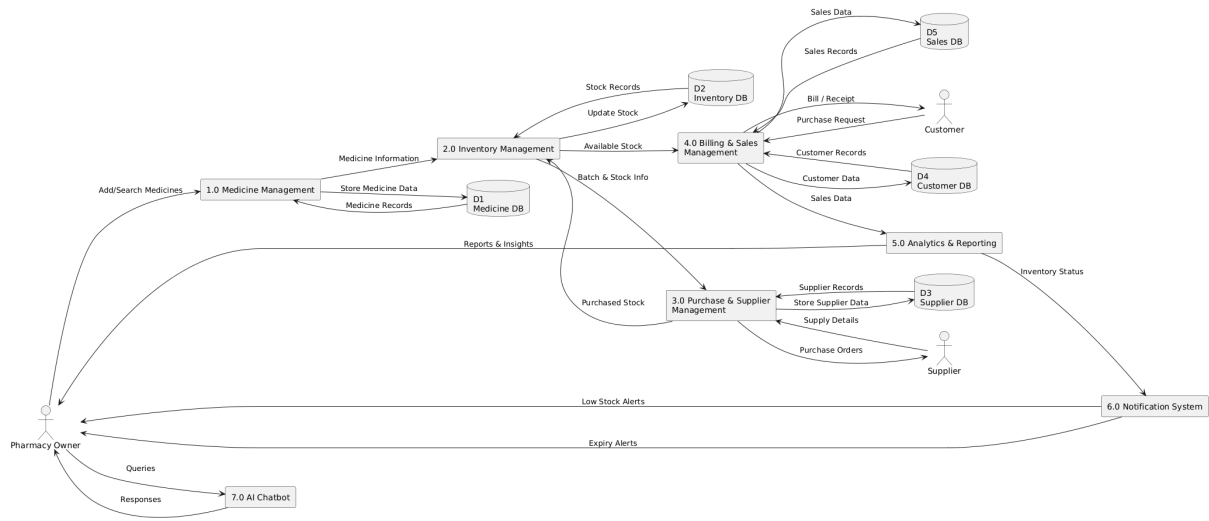


Figure 3.3: PharmaFlowAI Operational Activity Diagram

The diagram includes decision points such as whether a purchase is required and whether an alert needs to be generated.

3.3.3 Data Flow Diagram

The Data Flow Diagram represents how information moves between major system processes, data stores, and external actors.

The major processes include:

- Medicine Management.
- Inventory Management.
- Purchase and Supplier Management.
- Billing and Sales Management.
- Analytics and Reporting.
- Notification System.
- AI Chatbot.

The DFD shows the flow of medicine information, stock records, purchase requests, customer data, sales data, reports, alerts, and chatbot queries.

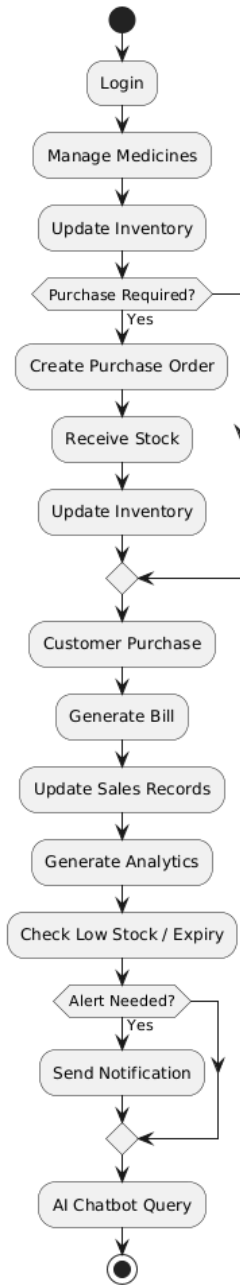


Figure 3.4: PharmaFlowAI Data Flow Diagram

3.4 Database Design

The database is organized around the major operational entities.

3.4.1 Medicine and Batch

Medicine represents a product, while batch information represents individual stock instances.

Batch-level information is useful because different batches of the same medicine may have different quantities and expiry dates.

3.4.2 Supplier and Purchase

Supplier records identify procurement sources.

Purchase records represent medicines acquired from suppliers and provide a connection to inventory.

3.4.3 Customer and Sales

Customer records contain information associated with pharmacy customers.

Sales represent transactions, while sale items represent individual medicines included in a transaction.

3.4.4 Payments

Payment information can be associated with sales and provides transaction-related financial information.

3.4.5 Notifications

Notifications represent operational conditions requiring attention, including low stock and expiry-related conditions.

3.4.6 Analytics

Analytics uses operational records to produce higher-level summaries of pharmacy activity.

3.5 Operational Data Flow

The simplified operational data flow is:

Medicine → Batch / Inventory → Purchase / Supplier → Billing / Sales → Analytics

Customer information participates in the sales process, while inventory conditions contribute to notifications.

3.6 Security Design

Authentication provides controlled entry into the system.

Data validation is also important for maintaining consistent application records.

Future security improvements may include detailed access-control review, deployment hardening, vulnerability assessment, and penetration testing.

3.7 AI and ML Design Considerations

The intelligent layer is designed as an extension of the operational platform.

Potential ML applications include:

- Demand forecasting.
- Stock-out risk prediction.
- Expiry-risk assessment.
- Reorder recommendations.

The final algorithm should be selected after the project dataset and prediction task have been clearly defined.

3.8 Design Principles

The main design principles are:

3.8.1 Modularity

Major pharmacy activities are separated into defined functional modules.

3.8.2 Data Consistency

Related activities use connected application entities.

3.8.3 Extensibility

The architecture allows future intelligent functionality to be added.

3.8.4 Usability

Common operations are accessible through the main workspace.

3.8.5 Maintainability

The Django structure separates major application responsibilities.

3.9 Design-to-Implementation Mapping

Design Component	Implementation Area
Authentication	Authentication and workspace
Medicine	Medicine catalogue
Batch and Inventory	Inventory management
Supplier and Purchase	Supplier and purchase management
Customer	Customer management
Sale and SaleItem	Billing and sales
Notification	Notification system
Analytics	Analytics dashboard
Prediction / Recommendation	Future ML functionality
Chatbot	AI chatbot prototype

Table 3.1: Design Components and Corresponding Implementation Areas

4 Implementation Phase

4.1 Implementation Overview

The core functionality of PharmaFlowAI is implemented using Django's modular architecture.

Django models and ORM manage application entities, application logic processes requests, and Django Templates provide the user interface.

The implementation follows the system structure defined during analysis and design.

4.2 Authentication Implementation

Authentication provides the initial entry point to the application.

The user provides the required credentials and, after successful authentication, gains access to the pharmacy workspace.

4.3 Dashboard Implementation

The dashboard provides an overview of important pharmacy information.

It acts as the main navigation point for accessing medicine, inventory, suppliers, customers, billing, notifications, and analytics.

4.4 Medicine Catalogue Implementation

The medicine catalogue maintains structured medicine records.

The module supports searching, filtering, and sorting.

Medicine information includes:

- Medicine name.
- Generic name.
- Strength.

- Category.
- Manufacturer.
- Unit.
- Prescription requirement.
- Active status.

The catalogue forms the product-level foundation for inventory and transactions.

4.5 Inventory and Batch Management

Inventory management focuses on available stock and batch-level information.

The module provides quantity, reorder level, status, shelf information, and expiry-related information.

Batch tracking allows stock to be considered at a more detailed level than a single aggregate quantity.

4.6 Supplier and Purchase Management

Supplier management provides a structured method for maintaining supplier information.

Purchase management records procurement activity and connects supplier information with medicines and incoming stock.

This module represents an important part of the inventory lifecycle.

4.7 Customer Management

The customer module maintains information associated with pharmacy customers.

Customer information can be used during billing and can also contribute to descriptive analytics such as returning-customer information.

4.8 Billing and Sales

The billing module records sales-related information.

A billing record can contain:

- Invoice number.
- Customer.

- Items.
- Amount.
- Transaction status.

Sales information provides both transaction records and historical information for analytics.

4.9 Notifications

The notification module makes important operational conditions visible.

Examples include low-stock conditions and approaching expiry.

The purpose is to direct the user's attention toward inventory that may require action.

4.10 Analytics

The analytics module provides descriptive information based on operational records.

Current indicators include:

- Sales this month.
- Sales growth.
- Average bill value.
- Top category.
- Returning customers.

The analytics layer provides the current descriptive-analysis capability of the project.

4.11 AI Chatbot Prototype

The AI chatbot has been developed as a conversational prototype.

The intended purpose is to allow users to ask pharmacy-related operational questions through a conversational interface.

Complete integration with operational application data is still pending.

Therefore, the current chatbot is treated as a prototype rather than a complete database-backed assistant.

4.12 Machine Learning Research

The ML component is currently focused on research, data generation, and data preparation.

Potential applications include:

- Demand forecasting.
- Stock-out prediction.
- Expiry-risk assessment.
- Reorder recommendations.

The project does not report model accuracy at this stage because the final dataset, prediction task, and model selection are still being established.

4.13 Django Admin

Django Admin provides administrative management of model-based application records.

It is useful during development for inspecting and maintaining structured data.

4.14 Module Integration

The major modules are connected through shared application data.

- Medicines connect with inventory and batches.
- Purchases connect suppliers with incoming stock.
- Sales connect customers with sold items.
- Transaction information contributes to analytics.
- Inventory conditions contribute to notifications.

This integration forms the central design principle of PharmaFlowAI.

4.15 Validation and Error Handling

Data-entry interfaces should validate required fields and maintain meaningful records.

Examples include validating medicine information, quantities, customer information, and billing fields.

Additional exception handling and validation rules can be introduced as the application evolves.

4.16 Maintainability

The Django structure separates models, application logic, and templates.

This makes individual modules easier to modify and provides a foundation for future additions such as ML services and deeper chatbot integration.

4.17 Implementation Workflow

The implementation workflow can be summarized as:

1. Define application entities.
2. Implement Django models.
3. Develop views and application logic.
4. Develop templates and interfaces.
5. Connect operational modules.
6. Add notifications and analytics.
7. Develop the chatbot prototype.
8. Generate and prepare data for future ML work.

5 Results and Evaluation

5.1 Testing Methodology

Testing focused on functional behavior, module integration, interface workflows, and the current state of intelligent functionality.

5.1.1 Unit Testing

Individual functionalities such as authentication, medicine management, inventory, billing, supplier management, and customer management were considered independently.

5.1.2 Integration Testing

Integration testing focused on relationships between purchasing, inventory, billing, customers, notifications, and analytics.

5.1.3 Interface Testing

The main workflows were reviewed through the application interface.

These included login, navigation, medicine search, inventory viewing, supplier management, billing, and analytics.

5.1.4 Performance Testing

The application was observed during normal usage for general responsiveness and stability.

Formal load and stress testing have not yet been completed.

5.1.5 AI Feature Testing

The chatbot prototype was tested using representative pharmacy-related questions.

The testing also highlighted the need for complete database integration for more advanced operational queries.

5.2 Representative Test Scenarios

Module	Scenario	Expected Result
Authentication	Enter valid credentials	Workspace is accessible
Medicine	Add medicine information	Medicine record is created
Medicine	Search and filter catalogue	Relevant records are displayed
Inventory	View stock information	Inventory status is visible
Batch	View batch information	Batch information is available
Supplier	Add supplier information	Supplier record is maintained
Purchase	Record purchase information	Purchase information is stored
Customer	Add customer information	Customer record is maintained
Billing	Create billing record	Transaction information is recorded
Notifications	Inspect attention conditions	Relevant conditions are displayed
Analytics	Open analytics dashboard	Operational summaries are displayed
Chatbot	Ask supported question	Prototype provides an available response

5.3 Functional Evaluation

The current implementation provides the major pharmacy management functions defined in the project scope.

The system provides a centralized workspace and separates operational areas into dedicated modules.

The implemented modules demonstrate the feasibility of the proposed system architecture.

5.4 Current Evaluation Status

Evaluation Area	Status
Authentication and Workspace	Implemented
Medicine Catalogue	Implemented
Inventory and Batch Management	Implemented
Supplier and Purchase Management	Implemented
Customer Management	Implemented
Billing and Sales	Implemented
Notifications	Implemented
Analytics Dashboard	Implemented
AI Chatbot Prototype	Developed
ML-based Analytics	Under Research
Formal Load Testing	Planned

Table 5.2: Current System Evaluation Summary

5.5 Observed System Results

The current implementation demonstrates the major operational workflow of the pharmacy management system.

The dashboard provides an overview while detailed modules support medicine, inventory, supplier, customer, and billing operations.

The analytics interface demonstrates the ability to summarize operational information.

In the demonstrated project state, the analytics interface displayed monthly sales of Rs. 84,260 together with other summary indicators.

This value represents project-interface data and is not presented as an externally validated business statistic.

5.6 Module Integration Analysis

The main benefit of the system is the connection between operational modules.

Purchasing represents incoming stock and is therefore related to inventory.

Billing represents outgoing sales and is connected to customer and sale information.

Transaction information can subsequently contribute to analytics.

Inventory conditions can also contribute to notifications.

This connected structure creates the data foundation required for future intelligent functionality.

5.7 AI Chatbot Evaluation

The chatbot prototype demonstrates a conversational approach to pharmacy information.

It provides an initial interface through which users can ask questions.

However, more advanced operational queries require reliable database integration.

Therefore, the current implementation should be considered an intermediate prototype rather than a fully integrated intelligent assistant.

5.8 Analytics Evaluation

The analytics module provides descriptive information derived from operational records.

Descriptive analytics helps the user understand current and historical activity.

Future predictive analytics can extend this capability by estimating likely future conditions rather than only summarizing past information.

5.9 Data Readiness for ML

Future ML datasets can include:

- Medicine identifiers.
- Transaction dates.
- Quantities sold.
- Purchase quantities.
- Inventory levels.
- Reorder levels.
- Batch and expiry information.
- Customer transaction frequency.

The final dataset and prediction target should be established before model training.

5.10 Usability Evaluation

The modular interface provides a logical separation of pharmacy activities.

Search and filtering are useful for locating medicine records.

Dashboard summaries provide an overview, while notifications can direct attention to inventory conditions.

5.11 Limitations

The current project has the following limitations:

1. ML model selection is still under research.
2. Historical project data is still being generated and prepared.
3. The chatbot is not completely connected to operational data.
4. Formal load and stress testing have not yet been completed.
5. Detailed deployment-level security testing remains future work.

These limitations describe the current development stage and do not prevent the implemented management modules from functioning as intended.

5.12 Overall Evaluation

The current implementation demonstrates the feasibility of a centralized Django-based pharmacy management platform.

The operational modules provide the primary management functionality, while notifications and analytics extend the system beyond basic record storage.

The chatbot provides an initial intelligent interaction layer, while ML research provides a direction for future prediction and recommendation functionality.

6 Conclusion and Future Work

6.1 Summary of Work

PharmaFlowAI was developed as a centralized web-based retail pharmacy management system using Python, Django, Django Templates, HTML, and CSS.

The platform combines medicine management, inventory, suppliers, purchases, customers, billing, sales, notifications, and analytics.

The project follows a modular Software Engineering approach in which individual functional areas are connected through shared application data.

The project also establishes a foundation for intelligent features through the AI chatbot prototype and ongoing Machine Learning research.

6.2 Key Findings

The main findings are:

- A centralized Django application can connect major pharmacy operations.
- Batch-level inventory provides a useful basis for stock and expiry monitoring.
- Connected purchasing, inventory, billing, and customer records support analytics.
- Notifications can help identify important operational conditions.
- A chatbot can provide a conversational interface but requires database integration for richer responses.
- ML-based analytics depends on sufficient and appropriately prepared historical data.

6.3 Future Work

6.3.1 Demand Forecasting

Historical sales records can be analyzed to identify demand patterns and estimate future medicine requirements.

6.3.2 Stock-Out Prediction

Future ML models can potentially identify medicines that are at risk of becoming unavailable based on historical demand and current inventory.

6.3.3 Expiry-Risk Assessment

Batch expiry information combined with sales patterns can potentially be used to identify inventory that requires attention before expiry.

6.3.4 Intelligent Reorder Recommendations

Future recommendations can combine stock quantity, reorder levels, historical demand, and supplier information to suggest procurement actions.

6.3.5 Chatbot Integration

The chatbot can be connected to operational application data through controlled application logic so that answers are based on verified pharmacy records.

6.3.6 Advanced Analytics

Future versions can include additional trends, reports, visualizations, and predictive indicators.

6.3.7 Security Improvements

Future development should include detailed security testing, access-control review, deployment hardening, and vulnerability assessment.

6.3.8 Scalability

The system can be extended to support larger datasets, multiple users, and potentially multiple pharmacy branches.

6.4 Recommendations

The following recommendations are proposed:

1. Continue generating consistent project data.
2. Define the exact ML prediction task before selecting an algorithm.
3. Establish a repeatable data-preparation pipeline.
4. Compare candidate algorithms using appropriate evaluation metrics.
5. Integrate the chatbot with verified operational data.
6. Perform formal performance and security testing before deployment.

6.5 Final Remarks

PharmaFlowAI demonstrates a centralized pharmacy management platform that combines operational management with a foundation for future intelligent decision support.

The current implementation provides the main functionality required for retail pharmacy operations.

The architecture, UML models, activity flow, and data flow provide a structured Software Engineering representation of the system.

The chatbot and ML components provide clear opportunities for further development as more project data becomes available.

A Requirements Traceability

The following table maps the major functional requirements to the corresponding system components.

ID	Requirement	Component
FR-01	User authentication	Authentication module
FR-02	Medicine management	Medicine catalogue
FR-03	Medicine search and filtering	Medicine catalogue
FR-04	Batch-level inventory	Inventory module
FR-05	Stock monitoring	Inventory and notifications
FR-06	Supplier management	Supplier module
FR-07	Purchase management	Purchase module
FR-08	Customer records	Customer module
FR-09	Billing and sales	Billing module
FR-10	Notifications	Notification module
FR-11	Analytics	Analytics module
FR-12	Conversational assistance	AI chatbot prototype
FR-13	Predictive analytics	Future ML module

A.1 Requirement Status

The major operational requirements are implemented in the current prototype.

The chatbot is developed as a prototype, while ML-based predictive functionality remains under research and future development.

B Testing Matrix

Test ID	Test Area	Purpose
T-01	Authentication	Verify workspace access
T-02	Medicine Creation	Verify medicine entry
T-03	Medicine Search	Verify retrieval and filtering
T-04	Inventory	Verify stock information
T-05	Batch Information	Verify batch records
T-06	Supplier	Verify supplier management
T-07	Purchase	Verify procurement records
T-08	Customer	Verify customer records
T-09	Billing	Verify transaction information
T-10	Notifications	Verify attention conditions
T-11	Analytics	Verify operational summaries
T-12	Chatbot	Evaluate prototype interaction
T-13	Integration	Verify module relationships
T-14	Performance	Observe general responsiveness

B.1 Testing Limitations

The current testing process is primarily functional and integration-oriented.

Formal automated coverage, stress testing, penetration testing, and production-scale benchmarking can be performed during future development.

C Project Module Summary

Module	Primary Responsibility
Authentication	Controlled workspace access
Dashboard	High-level system overview
Medicine	Catalogue and product information
Inventory	Stock and batch information
Supplier	Supplier information
Purchase	Procurement records
Customer	Customer information
Billing	Sales and invoice records
Notifications	Operational attention conditions
Analytics	Descriptive operational summaries
AI Chatbot	Conversational prototype
ML Research	Future predictive capabilities